

Lecture 5 Gradient Descent via Quadratic Constraints

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In Lecture 3, we modeled optimization methods as feedback systems and used quadratic constraints and dissipation inequalities to certify convergence. This lecture develops that analysis for the gradient method applied to the unconstrained minimization problem

$$\min_{x \in \mathbb{R}^p} f(x), \quad (5.1)$$

where $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is differentiable, L -smooth, and m -strongly convex, with $0 < m \leq L$. A point $x^* \in \mathbb{R}^p$ is a global minimizer if $f(x^*) \leq f(x)$ for every $x \in \mathbb{R}^p$.

We will answer the following three questions:

1. Does a global minimizer x^* exist, and is it unique?
2. What algorithm can we use to find x^* ?
3. How do the stepsize and the constants (m, L) determine its convergence rate?

Throughout this lecture, $\|x\|$ denotes the Euclidean norm, $I = I_p$ denotes the $p \times p$ identity matrix, and matrix inequalities $\preceq$ and $\succeq$ refer to the semidefinite order.

5.1 Some definitions

A differentiable function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is L -smooth if, for all $x, y \in \mathbb{R}^p$,

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|. \quad (5.2)$$

Smoothness implies the following useful inequality:

$$f(x) \leq f(y) + \nabla f(y)^\top (x - y) + \frac{L}{2}\|x - y\|^2. \quad (5.3)$$

One can prove (5.3) by integrating the gradient along the line segment from y to x ; this is a useful exercise.

We say f is m -strongly convex, for some $m > 0$, if, for all $x, y \in \mathbb{R}^p$,

$$f(x) \geq f(y) + \nabla f(y)^\top (x - y) + \frac{m}{2}\|x - y\|^2. \quad (5.4)$$

For any fixed y , the right side of (5.3) is a quadratic upper bound for f , and the right side of (5.4) is a quadratic lower bound. These bounds explain why smooth strongly convex

functions are a useful starting point for optimization analysis. Comparing them also shows that $m \leq L$.

If f is L -smooth and m -strongly convex, the following inequality, often called the strong co-coercivity inequality, holds for all $x, y \in \mathbb{R}^p$:

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \frac{mL}{m+L} \|x - y\|^2 + \frac{1}{m+L} \|\nabla f(x) - \nabla f(y)\|^2. \quad (5.5)$$

Its proof is another useful exercise. This inequality is equivalent to the quadratic constraint

$$\begin{bmatrix} x - y \\ \nabla f(x) - \nabla f(y) \end{bmatrix}^\top \begin{bmatrix} -2mLI & (m+L)I \\ (m+L)I & -2I \end{bmatrix} \begin{bmatrix} x - y \\ \nabla f(x) - \nabla f(y) \end{bmatrix} \geq 0. \quad (5.6)$$

Writing $d = x - y$ and $g = \nabla f(x) - \nabla f(y)$, the same condition is $(Ld - g)^\top (g - md) \geq 0$. This is precisely the sector bound used in Lecture 3. The matrix in (5.6) is the negative of the matrix used there for a quadratic constraint with a nonpositive sign.

The constraint holds for pairs (d, g) generated by the gradient of f . It does *not* assert that the matrix in (5.6) is positive semidefinite on all of $\mathbb{R}^{2p}$.

5.2 Answer 1: existence and uniqueness

The existence of x^* follows from the next theorem, a consequence of the extreme value theorem.

Theorem 5.1. *If a continuous function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ satisfies $f(x) \rightarrow +\infty$ as $\|x\| \rightarrow \infty$, then f has a global minimizer.*

For a strongly convex function, fix $y = 0$ in (5.4):

$$\begin{aligned} f(x) &\geq f(0) + \nabla f(0)^\top x + \frac{m}{2} \|x\|^2 \\ &\geq f(0) - \|\nabla f(0)\| \|x\| + \frac{m}{2} \|x\|^2. \end{aligned}$$

The last expression tends to $+\infty$ as $\|x\| \rightarrow \infty$, so a global minimizer exists.

Another useful theorem is the first-order optimality condition.

Theorem 5.2. *If f is differentiable and x^* is an unconstrained global minimizer, then $\nabla f(x^*) = 0$. If f is also convex, this condition is sufficient for global optimality.*

For necessity, if $\nabla f(x^*) \neq 0$, the directional derivative in the direction $-\nabla f(x^*)$ is negative, contradicting minimality. Sufficiency follows from the first-order inequality for convexity.

To prove uniqueness, assume x_1^* and x_2^* are both global minimizers. Since $\nabla f(x_1^*) = \nabla f(x_2^*) = 0$, (5.4) gives

$$\begin{aligned} f(x_1^*) &\geq f(x_2^*) + \frac{m}{2} \|x_1^* - x_2^*\|^2, \\ f(x_2^*) &\geq f(x_1^*) + \frac{m}{2} \|x_1^* - x_2^*\|^2. \end{aligned}$$

Adding these inequalities yields $m\|x_1^* - x_2^*\|^2 \leq 0$. Thus $x_1^* = x_2^*$, and the global minimizer is unique. The L -smoothness assumption is not needed for this existence and uniqueness argument.

5.3 Answer 2: the gradient method

The gradient descent method iterates as

$$x_{k+1} = x_k - \alpha \nabla f(x_k), \quad (5.7)$$

where $\alpha > 0$ is a constant stepsize. Starting from any $x_0 \in \mathbb{R}^p$, we compute $x_1, x_2, \dots$ using only gradient evaluations. Thus gradient descent is a first-order optimization method. With a suitable stepsize, it converges linearly for smooth strongly convex objectives.

As in Lecture 3, (5.7) is the feedback interconnection

$$\begin{aligned} \xi_{k+1} &= A\xi_k + Bw_k, \\ v_k &= C\xi_k, \quad A = I, \quad B = -\alpha I, \quad C = I, \quad \xi_k = x_k. \\ w_k &= \nabla f(v_k), \end{aligned} \quad (5.8)$$

The equilibrium is $(\xi^*, v^*, w^*) = (x^*, x^*, 0)$. Define the error $e_k = x_k - x^*$. Then

$$e_{k+1} = e_k - \alpha w_k, \quad w_k = \nabla f(x_k). \quad (5.9)$$

Setting $y = x^*$ in (5.6) and using $\nabla f(x^*) = 0$ provides the quadratic constraint on (e_k, w_k) needed below.

5.4 Answer 3: linear convergence

We seek a bound of the form

$$\|x_k - x^*\| \leq \rho^k \|x_0 - x^*\| \quad (5.10)$$

with $0 \leq \rho < 1$. For $0 < \rho < 1$, this is called exponential convergence in control and linear convergence in optimization: the logarithm of the bound decreases linearly with k . A smaller ρ gives a faster certified rate. The case $\rho = 0$ means that $x_1 = x^*$ and all later iterates stay at x^* .

The following theorem describes the dependence between ρ and (α, m, L) .

Theorem 5.3. Suppose f is L -smooth and m -strongly convex, where $0 < m \leq L$, and let x^* be its unique global minimizer. Given a stepsize $\alpha > 0$, if there exist $0 \leq \rho < 1$ and $\lambda \geq 0$ such that

$$H(\alpha, \rho, \lambda) := \begin{bmatrix} 1 - \rho^2 & -\alpha \\ -\alpha & \alpha^2 \end{bmatrix} + \lambda \begin{bmatrix} -2mL & m + L \\ m + L & -2 \end{bmatrix} \preceq 0, \quad (5.11)$$

then the gradient method satisfies (5.10) for every x_0 and every $k \geq 1$.

The condition (5.11) is a sufficient certificate of convergence. For a fixed stepsize α , it is linear in the decision variables ρ^2 and λ . It is therefore a small semidefinite program. We will obtain explicit certificates by choosing these scalars analytically.

5.4.1 A useful dissipation lemma

The following lemma isolates the argument that converts a quadratic constraint into a rate bound.

Lemma 5.4. Suppose $e_k, w_k \in \mathbb{R}^p$ satisfy $e_{k+1} = e_k - \alpha w_k$. Let $M = M^\top \in \mathbb{R}^{2p \times 2p}$ satisfy

$$\begin{bmatrix} e_k \\ w_k \end{bmatrix}^\top M \begin{bmatrix} e_k \\ w_k \end{bmatrix} \geq 0 \quad (5.12)$$

for every k . If there exist $0 \leq \rho < 1$ and $\lambda \geq 0$ such that

$$\begin{bmatrix} (1 - \rho^2)I & -\alpha I \\ -\alpha I & \alpha^2 I \end{bmatrix} + \lambda M \preceq 0, \quad (5.13)$$

then $\|e_k\| \leq \rho^k \|e_0\|$ for every $k \geq 1$.

Proof: Choose the storage function $V(e) = \|e\|^2$. The linear update gives

$$\begin{aligned} V(e_{k+1}) - \rho^2 V(e_k) &= (1 - \rho^2) \|e_k\|^2 - 2\alpha e_k^\top w_k + \alpha^2 \|w_k\|^2 \\ &= \begin{bmatrix} e_k \\ w_k \end{bmatrix}^\top \begin{bmatrix} (1 - \rho^2)I & -\alpha I \\ -\alpha I & \alpha^2 I \end{bmatrix} \begin{bmatrix} e_k \\ w_k \end{bmatrix}. \end{aligned} \quad (5.14)$$

Multiplying (5.13) on the left and right by $\begin{bmatrix} e_k \\ w_k \end{bmatrix}^\top$ and $\begin{bmatrix} e_k \\ w_k \end{bmatrix}$, respectively, yields

$$V(e_{k+1}) - \rho^2 V(e_k) \leq -\lambda \begin{bmatrix} e_k \\ w_k \end{bmatrix}^\top M \begin{bmatrix} e_k \\ w_k \end{bmatrix} \leq 0. \quad (5.15)$$

The final inequality uses (5.12) and $\lambda \geq 0$. Iterating $V(e_{k+1}) \leq \rho^2 V(e_k)$ and taking square roots proves the claim. ■

Equation (5.15) is the exponential dissipation inequality from Lecture 3, with supply rate $S(e_k, w_k) = -\lambda \begin{bmatrix} e_k \\ w_k \end{bmatrix}^\top M \begin{bmatrix} e_k \\ w_k \end{bmatrix} \leq 0$. The quadratic constraint is only required along the admissible pairs (e_k, w_k) ; M itself need not be positive semidefinite.

5.4.2 Proof of the gradient convergence theorem

Proof (Theorem 5.3): Take $e_k = x_k - x^*$ and $w_k = \nabla f(x_k)$. The gradient method gives (5.9), and (5.6) with $y = x^*$ gives

$$\begin{bmatrix} e_k \\ w_k \end{bmatrix}^\top \underbrace{\begin{bmatrix} -2mLI & (m+L)I \\ (m+L)I & -2I \end{bmatrix}}_M \begin{bmatrix} e_k \\ w_k \end{bmatrix} \geq 0. \quad (5.16)$$

The scalar matrix condition (5.11) is equivalent to (5.13) with this choice of M . Indeed, if $H = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \preceq 0$, then for any $e, w \in \mathbb{R}^p$,

$$\begin{bmatrix} e \\ w \end{bmatrix}^\top \begin{bmatrix} aI & bI \\ bI & cI \end{bmatrix} \begin{bmatrix} e \\ w \end{bmatrix} = \sum_{i=1}^p \begin{bmatrix} e_i \\ w_i \end{bmatrix}^\top H \begin{bmatrix} e_i \\ w_i \end{bmatrix} \leq 0.$$

The converse follows by choosing vectors supported on one coordinate. Lemma 5.4 now gives $\|x_k - x^*\| \leq \rho^k \|x_0 - x^*\|$. ■

This proof separates the analysis into the linear update, the quadratic constraint on the gradient, and a matrix inequality combining the two. The same construction is the basis of the more general feedback-system analysis in Lecture 3.

5.5 Stepsize choices and convergence rates

We now apply Theorem 5.3 to the two stepsizes discussed in Lecture 3. First recall that a symmetric matrix A is positive semidefinite if $z^\top A z \geq 0$ for every z , and negative semidefinite if $z^\top A z \leq 0$ for every z . A symmetric 2×2 matrix satisfies

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \preceq 0 \iff a \leq 0, \quad c \leq 0, \quad ac - b^2 \geq 0. \quad (5.17)$$

All three conditions are needed, including when a diagonal entry is zero. Another useful way to establish semidefiniteness is a factorization: for example,

$$\begin{bmatrix} -m^2 & m \\ m & -1 \end{bmatrix} = - \begin{bmatrix} m \\ -1 \end{bmatrix} \begin{bmatrix} m \\ -1 \end{bmatrix}^\top \preceq 0.$$

5.5.1 Case 1: $\alpha = 1/L$

Choose

$$\alpha = \frac{1}{L}, \quad \rho = 1 - \frac{m}{L}, \quad \lambda = \frac{1}{L^2}.$$

Substitution into (5.11) gives

$$H\left(\frac{1}{L}, 1 - \frac{m}{L}, \frac{1}{L^2}\right) = \frac{1}{L^2} \begin{bmatrix} -m^2 & m \\ m & -1 \end{bmatrix} \preceq 0. \quad (5.18)$$

Hence

$$\|x_k - x^*\| \leq \left(1 - \frac{m}{L}\right)^k \|x_0 - x^*\|. \quad (5.19)$$

5.5.2 Case 2: $\alpha = 2/(m + L)$

Choose

$$\alpha = \frac{2}{m + L}, \quad \rho = \frac{L - m}{L + m}, \quad \lambda = \frac{2}{(m + L)^2}.$$

In this case every entry cancels:

$$H\left(\frac{2}{m + L}, \frac{L - m}{L + m}, \frac{2}{(m + L)^2}\right) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \preceq 0. \quad (5.20)$$

Therefore

$$\|x_k - x^*\| \leq \left(\frac{L - m}{L + m}\right)^k \|x_0 - x^*\|. \quad (5.21)$$

Since $0 < m \leq L$,

$$\frac{L - m}{L + m} \leq 1 - \frac{m}{L}.$$

Thus $\alpha = 2/(m + L)$ has the smaller certified contraction factor, strictly so when $m < L$. It requires both m and L , whereas $\alpha = 1/L$ only requires L to set the stepsize. If $m = L$, both choices coincide and the certificates give $\rho = 0$: the method reaches x^* in one step.

5.5.3 General constant stepsizes

To study an arbitrary $\alpha > 0$, expand the matrix in (5.11):

$$H(\alpha, \rho, \lambda) = \begin{bmatrix} 1 - \rho^2 - 2mL\lambda & \lambda(m+L) - \alpha \\ \lambda(m+L) - \alpha & \alpha^2 - 2\lambda \end{bmatrix}. \quad (5.22)$$

By (5.17), the exact conditions are

$$\begin{aligned} 1 - \rho^2 - 2mL\lambda &\leq 0, \\ \alpha^2 - 2\lambda &\leq 0, \\ (1 - \rho^2 - 2mL\lambda)(\alpha^2 - 2\lambda) - (\lambda(m+L) - \alpha)^2 &\geq 0. \end{aligned} \quad (5.23)$$

When $\lambda > \alpha^2/2$, the second diagonal entry is strictly negative. In this case the conditions reduce to

$$\rho^2 \geq 1 - 2mL\lambda + \frac{(\lambda(m+L) - \alpha)^2}{2\lambda - \alpha^2}. \quad (5.24)$$

This formula can be used to search for a multiplier λ giving a small rate ρ .

The boundary $\lambda = \alpha^2/2$ must be checked separately, since the denominator in (5.24) vanishes. Then (5.23) requires

$$\lambda(m+L) - \alpha = 0, \quad 1 - \rho^2 - 2mL\lambda \leq 0.$$

For $\alpha > 0$, these conditions give $\alpha = 2/(m+L)$ and $\rho^2 \geq (L-m)^2/(L+m)^2$, recovering Case 2.

Exercise: dependence on the stepsize. For $0 < m < L$ and $0 < \alpha < 2/L$, use (5.23) to verify that the rate

$$\rho(\alpha) = \max\{|1 - \alpha m|, |1 - \alpha L|\} \quad (5.25)$$

is certified by the multiplier

$$\lambda(\alpha) = \begin{cases} \frac{\alpha(1 - \alpha m)}{L - m}, & 0 < \alpha \leq \frac{2}{m+L}, \\ \frac{\alpha(\alpha L - 1)}{L - m}, & \frac{2}{m+L} \leq \alpha < \frac{2}{L}. \end{cases}$$

The two expressions agree at $\alpha = 2/(m+L)$. Show that this stepsize minimizes (5.25) and gives $\rho = (L-m)/(L+m)$. As a check, for the scalar quadratic $f(x) = \frac{q}{2}x^2$ with $q = m$ or $q = L$, the iteration is $x_{k+1} = (1 - \alpha q)x_k$. These examples attain the two factors in (5.25) and explain why the bound is sharp over the function class.